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CO1-AT-1-Short Answer Test

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1. What is Data Science? Explain its primary objective

A. Data Science is an interdisciplinary field that uses statistics, mathematics, programming, and machine learning to extract meaningful insights from data.

2. Why is Data Science important in today's digital world? Mention any four reasons.

* Helps in data-driven decision making.
* Improves business efficiency and productivity.
* Enables accurate predictions using historical data.
* Provides personalized services and recommendations.

3. List any four benefits of Data Science in business and Society.

* Better decision making.
* Fraud detection and risk management.
* improved customer satisfaction.
* Healthcare and disease Prediction.

4. Mention four real-world applications of Data Science.

* Healthcare (disease Prediction)
* Banking (fraud detection)
* E-commerce (Product recommendations)
* Transportation (traffic and route optimization).

5. What are the different facets of data? Briefly explain each.

* Volume: Amount of data generated.
* Velocity: Speed at which data is generated and processed.
* Variety: Different forms of data (text, images, videos, etc.)
* Veracity: Accuracy and reliability of data.

*Value: Usefulness of data for decision making.

6. Differentiate between structured, semi-structured, and unstructured data with suitable examples.

Structured :- organized in rows and colums ex:- MySQL Database

Semi-Structured :- Partially organized ex:- JSON, XML

Unstructured :- No fixed format ex:- Images, Videos, Emails

7. What is Big Data? State its key characteristics

Big Data refers to extremely large and complex datasets that cannot be processed using traditional methods.

Characteristics (5Vs) :- Volume, Velocity, Variety, Veracity, Value

8. Explain the Big Data ecosystem. Name any four technologies used in it.

The Big Data ecosystem consists of tools used for collecting, storing, processing, and analyzing large datasets.

Technologies:

* Hadoop
* Spark
* Hive
* HBase

9. What are the major Phases of the Data Science process?

* Data Collection
* Data Cleaning
* Data integration
* Data Transformation
* Data Analysis
* Model Building
* Visualization and Deployment.

10. What is data retrieval?. Mention two common sources from which data can be retrieved.

Data retrieval is the process of obtaining data from different storage systems.

Sources:

* Databases (SQL)
* Web APIs / Internet

11. Why is data collection considered an important step in the Data Science process?.

* Provides quality data for analysis.
* Improves prediction accuracy
* Reduces errors

12. What is data cleansing? List any four common data-Cleaning techniques.

Data cleansing is the process of identifying and correcting errors in data.

Techniques:

* Remove duplicate records.
* Handle missing values.

13. What is data integration! Why is it necessary in Data Science?.

Data integration is the process of combining data from multiple sources into a single dataset.

Necessity:

* improves data quality.
* Eliminates redundancy.

14 Explain data transformation with suitable examples

Data transformation is the process of converting data into a suitable format for analysis.

Examples:

* Converting text into numbers
* Normalizing data values.

15 What is data analysis? Mention any four techniques used for data analysis.

Data analysis is the process of examining data to discover useful information and support decision-making.

Techniques:

* Classification
* Clustering.

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